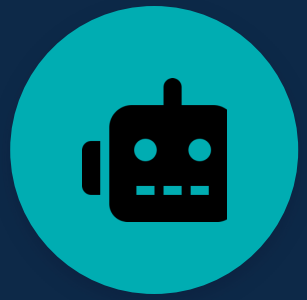
MICROSOFT FABRIC · PREVIEW FEATURE

# Data Agents
## in Microsoft Fabric



Ask your data a question in natural language, how it works, how to build one, and how it stacks up against everyone else's version.

**Samantha Cruz**

Data Engineering Team Lead at isolutions

Queensland Power BI and Fabric User Group · 6 August 2026

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# Quick hello

## I’m Samantha



Data Engineering Team Lead at isolutions

4× Microsoft + 4× Databricks certified

8+ years working with data across a bunch of different industries

Fabric and Databricks are what I use day to day, this talk comes from hands-on scars, not just documentation

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ROADMAP

# What we'll cover tonight



### What it is

The building blocks, in plain terms



### How it works

What happens between your question and the answer



### Build & activate

Creating one, and the setup everyone forgets



### Evaluate & consume

Checking it's right, and where you can use it



### How it compares

Copilot, and the rest of the industry



### Live demo

Let’s build an agent with real e-commerce data

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# The problem we're actually solving

**TODAY**

- Business user has a question
- Pings an analyst on Slack/Teams
- Analyst context-switches, writes SQL/DAX
- Answer arrives... Other tasks delayed
- Question #2 starts the whole loop again

**WITH A DATA AGENT**

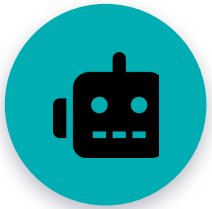
- Business user asks in plain English
- Agent checks permissions, same rules as always
- Translates the question to SQL/DAX/KQL
- Answer comes back in seconds
- No analyst needed for the routine stuff

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DEFINITIONS

STILL IN PREVIEW

# So, what actually is a Data Agent?



A Fabric capability that lets people have natural-language conversations with enterprise data, built on the Azure OpenAI Assistant API, without ever exposing SQL, DAX, or KQL to the person asking.



### Conversational

Plain-English questions in, structured answers out



### An abstraction layer

Sits over your query engines, doesn't replace them



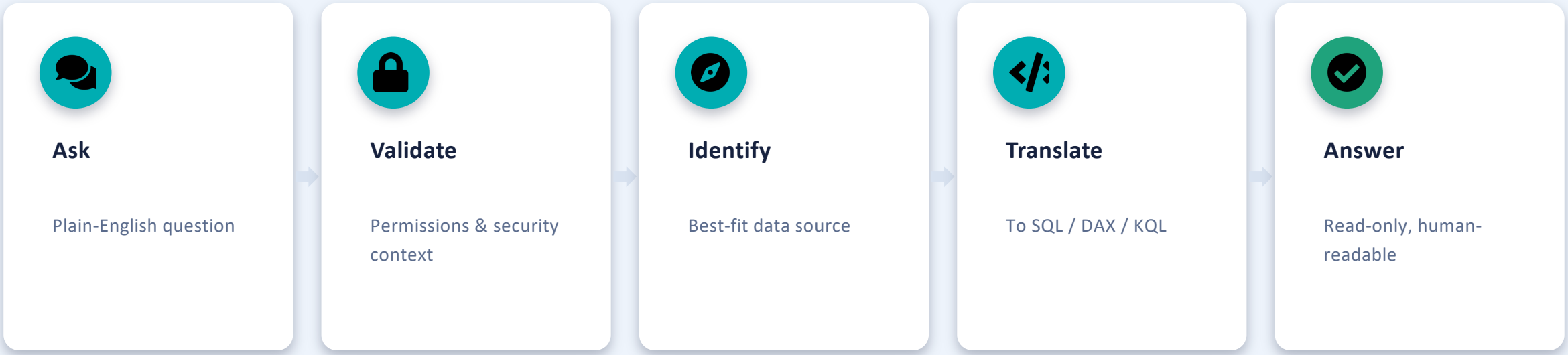
### Permission-aware

Inherits exactly the access you already have, no shortcuts

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UNDER THE HOOD

# From question to answer




*“If the underlying data model is messy, the agent's answers will be messy too.”*

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CONNECTIVITY

# One agent, up to five of six data sources

| Category | Artifacts | Query language | Key scenario |
|---|---|---|---|
| **SQL** | Lakehouse, Warehouse, SQL Database, Mirrored DBs | T-SQL | Structured analytics over relational & Delta Lake data |
| **Eventhouse** | Eventhouse KQL Database | KQL | Real-time intelligence & time-series analytics |
| **Semantic Model** | Power BI Semantic Models | DAX | Business logic, calculated measures, curated metrics |
| **Graph** *(preview)* | Fabric Graph Model | GQL | Relationship-rich data exploration & graph analytics |
| **Ontology** *(preview)* | Fabric Ontology | Ontology-native | Domain knowledge & semantic context |
| **Azure AI Search** *(preview)* | Azure AI Search Index | NL + search | Unstructured retrieval — PDFs, text, enriched docs |

Important: it doesn't copy or move any of this data. It reads what's already there, live, and it's strictly read-only. You can combine different data sources.

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BUILD IT

MIN. F2 CAPACITY

# Creating a Data Agent (no code required)

### 1. Define context

What is this agent for?
Domain, tone, scope

### 2. Select source(s)

Pick the Lakehouse /
Warehouse / semantic model

### 3. Add examples

Sample question → query pairs
for tricky logic

### 4. Test it

Ask it real questions before
you publish

*It's created as an artifact inside your Fabric workspace — new item, pick "Data Agent," name it, done. No notebooks, no code.*

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THE BIT EVERYONE FORGETS

# It also depends on tenant settings



Data Agents rely on tenant-wide settings, you need to enable Copilot / Azure OpenAI tenant-wide.



**Data residency**

Cross-geo processing settings can block it entirely



**Compliance**

security first



**IT conversation**

This is a tenant-admin decision, not a workspace one

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ONCE IT'S BUILT

# One agent, many ways to use it

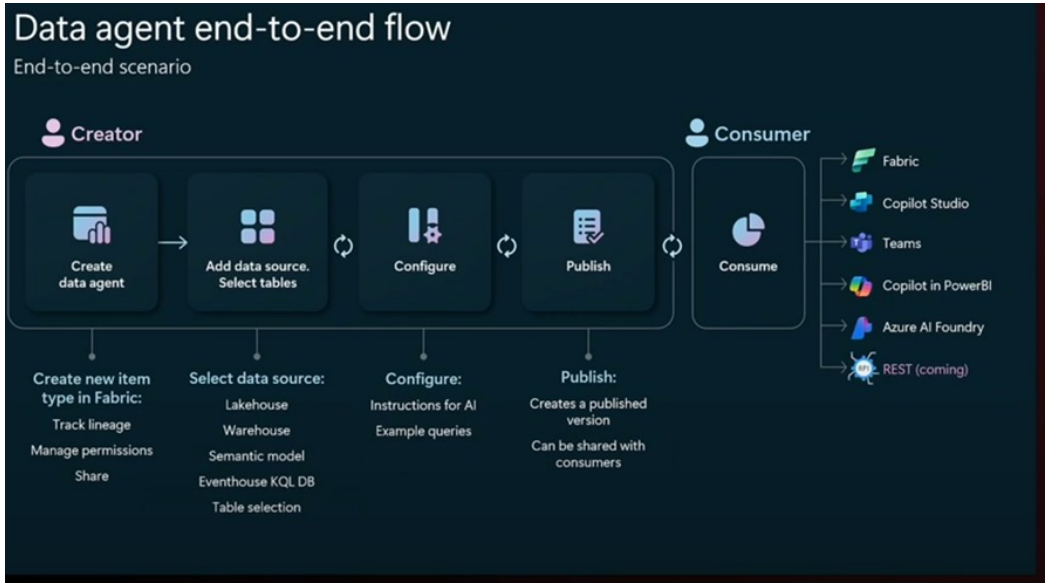
Copilot in Power BI

Microsoft Copilot Studio

Azure AI Foundry

Python SDK (programmatic)

Microsoft 365 Copilot




*One agent, published once, consumed everywhere it's needed.*

*Same agent, defined once, consumed interactively by people, or programmatically by other systems. Think of it as a data product.*

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TRUST, BUT VERIFY

# Yes, you can test an agent's answers

### Why bother?

Catch bad answers before your users do. Validate changes to config or the underlying data model. This is the difference between a toy and something you'd put in front of a client.

```
from fabric.dataagent.evaluation import *
eval_id = evaluate_data_agent(df, agent)
summary = get_evaluation_summary(eval_id)
```

| index | evaluation_id | true_count | false_count | unclear_count | true_percentage |
|---|---|---|---|---|---|
| 0 | 2cf897c6-801c-4e94-8664-f4848ffb6071 | 3 | 1 | 0 | 75.0 |
| 1 | 30c590da-6df8-494f-a359-642d34d4fa32 | 2 | 2 | 0 | 50.0 |
| 2 | 34fc1c4c-8696-48e5-aebb-78b3b681cc93 | 2 | 2 | 0 | 50.0 |
| 3 | 3879189e-f216-4f43-ae2e-9f96933a1429 | 2 | 2 | 0 | 50.0 |
| 4 | 45dfc4ae-a6ae-4265-80ff-a75c7b463fb0 | 4 | 0 | 0 | 100.0 |
| 5 | 726cb9fe-614a-4dc0-b121-18dd224aa391 | 2 | 2 | 0 | 50.0 |
| 6 | 75d8e7db-3448-4e93-bb20-819e038e28a6 | 4 | 0 | 0 | 100.0 |
| 7 | 760af511-c8a8-497c-b729-6b113f4ac2a9 | 3 | 1 | 0 | 75.0 |
| 8 | 76e1e1c8-29ee-4505-bce2-9cd331927ef4 | 3 | 1 | 0 | 75.0 |
| 9 | 7b65fc7b-c8c8-46a1-9c2b-04adf65f4f48 | 3 | 1 | 0 | 75.0 |

*Batch results: pass/fail counts per question set*

### Keep in mind

It's batch, not real-time. There's no GUI for it yet, it's Python-SDK only. And running an evaluation never changes the agent itself, it just reports on it.

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KEEPING IT HONEST

# What a Data Agent can't do (yet)

### It doesn't do this

No write: can't create, update, or delete anything

Can't generate charts or visualizations

No explicit SQL joins you can hand-tune

Limited control over the generated query

### Still your job

Data governance stays your team's job

Data quality: the agent won't fix bad data

Config still depends on tenant settings

Preview status: not yet built for critical workloads

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# A few things that make a huge difference



### Name things clearly

Clear table/column names help the agent



### One domain per agent

Specialize instead of building one giant do-everything agent



### Minimize scope

Only include the tables you need answered



### Say what TO do

Positive instructions beat a list of things to avoid



### Define your jargon

Spell out business terms, abbreviations, synonyms



### Use example queries

Show it the joins/filters that are too complex to guess

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# Data Agent vs. Copilot vs. Copilot Studio

| | Copilot in Fabric | Fabric Data Agent | Copilot Studio |
|---|---|---|---|
| **Scope** | Single-step, in-tool help | Multi-step, autonomous, standalone | Orchestrates many agents/tools |
| **Memory** | Session-bound | Keeps workspace context across runs | Depends on the orchestrator |
| **Best for** | Iterating inside a report | Recurring, governed Q&A | Cross-system scenarios (Teams, M365) |

Copilot answers: **“help me write this query.”** Data Agent answers: **“produce the monthly churn report using our lakehouse.”**

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ZOOMING OUT

NEW FOR 2026

# Every platform has built this now

| Platform | Their equivalent | How it works |
|---|---|---|
| **Microsoft Fabric** | **Data Agent** | NL → SQL/DAX/KQL, read-only, Purview-governed |
| **Databricks** | **Genie Spaces** | NL → SQL over Unity Catalog; new managed MCP server |
| **Snowflake** | **Cortex Agents / Snowflake Intelligence** | Fuses structured (Analyst) + unstructured (Search) |
| **Google Cloud** | **Conversational Analytics** | Gemini over BigQuery + Looker — GA since Jul 2026 |
| **AWS** | **Amazon Q in QuickSight** | NL prompt → full dashboard, deep AWS-native |
| **ThoughtSpot** | **Spotter agents** | NL search-to-insight, GA push in early 2026 |

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THE ACTUAL TAKEAWAY

# Same pattern, different governance backbone

### The mechanism converges

Ground an LLM in a governed semantic/catalog layer → NL to structured query → keep it read-only & permission-scoped.

### The moat is governance

Fabric leans on Purview + OneLake. Databricks leans on Unity Catalog. Snowflake fuses structured + unstructured in one agent.

### Pick by metadata, not hype

Choose the data agent that matches where your governed semantic layer already lives not the one with the flashiest demo.

***This isn't a “best agent wins” decision, it's a “who already owns my metadata” decision.***

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STAYING CURRENT

# What's shipped in 2026

| | | |
|---|---|---|
| MAR 2026 | **Fabric Remote MCP** | Cloud-hosted MCP server, external agents can act on Fabric via Entra ID + existing RBAC |
| JUN 2026 | **Service principals for Data Agents** | Apps authenticate with their own identity, no per-request user sign-in |
| JUN 2026 | **Data Agent Runtime (preview)** | Better default behavior, observability, reliability |
| JUL 2026 | **Fabric local MCP** | Open-source local server for coding assistants to manage Fabric |

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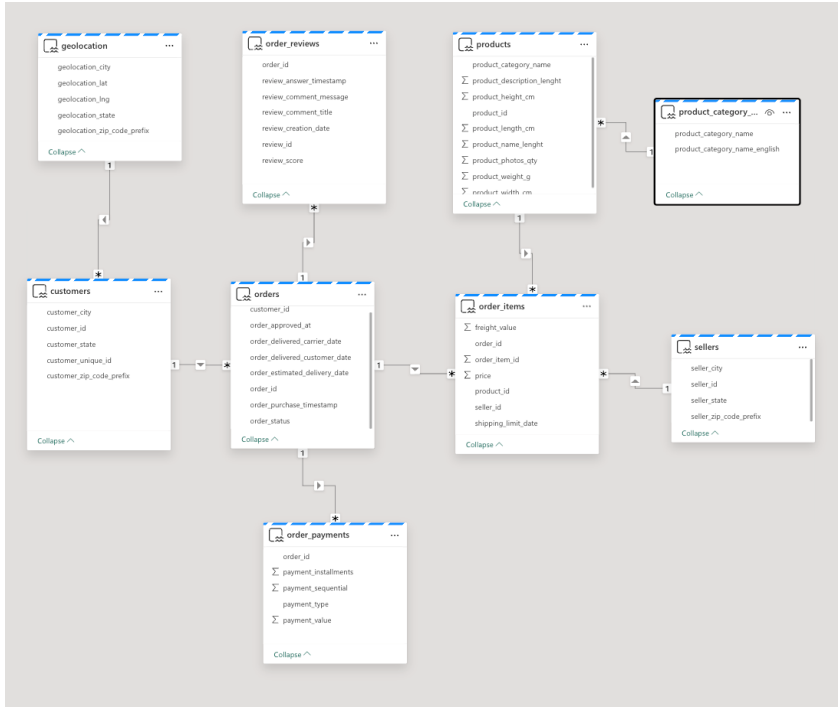


# Demo time

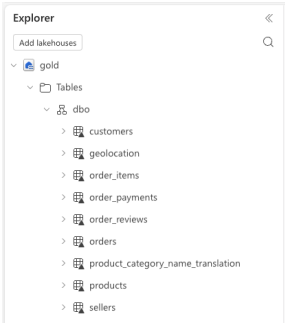
Let's see this in a real workspace

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# Real e-commerce data: Olist




*9 related tables: customers, orders, payments, reviews, products, sellers...*




*Gold layer, already modeled in the Lakehouse*

## Why this dataset works

Genuine relationships + clear business questions, exactly what a Data Agent needs to shine.

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WHAT YOU'RE ABOUT TO WATCH

# The live run-of-show



### 1. Show tables

The gold Lakehouse, already built, no data-building live



### 2. Build agent

Pick the source, add an instruction + example query



### 3. Ask questions

Increasing difficulty including one needing our example



### 4. Poke limits

Try to make it write something, watch it refuse



### 5. Evaluate

The Python SDK from earlier, for real this time

*Sample questions: “Average review score by product category?” · “Which sellers had the most delayed deliveries?”*

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# Thank you!

Questions, war stories, or “that didn't work when I tried it” all welcome.

Samantha Cruz · Data Engineering Team Lead, isolutions

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